BIOL 40C Cumulative Study Sheet

Weeks 0–4 · Spring 2026 · For Midterm 2 (Wed May 13)

> **How to use this sheet.** Cover the right column. Read each topic on the left. Recite what you can. Then check yourself. The act of trying to recall is what builds memory; rereading does not. If you can hit ~80% of the right column from memory, you are ready.

Unit 1 · Digestive System (Weeks 0–1)

Four layers of the GI tract wall	Mucosa (epithelium + lamina propria + muscularis mucosae) → submucosa (with Meissner's plexus, glands) → muscularis externa (with Auerbach's plexus, peristalsis) → serosa.
Stomach cell types and what they secrete	**Parietal:** HCl + intrinsic factor (B12 absorption). **Chief:** pepsinogen. **G cells:** gastrin. **Mucous neck:** mucus + bicarb (protection).
How parietal cells make HCl	H^+/K^+ ATPase pumps H^+ into the lumen (this is the secretory pump, not the basolateral exchanger). Cl^- follows. Bicarbonate goes the other way (the "alkaline tide").
Three regions of the small intestine and their jobs	**Duodenum:** chemical digestion (pancreatic enzymes, bile arrive here). **Jejunum:** the workhorse of absorption. **Ileum:** B12 + bile salt reabsorption.
Why the small intestine has so much surface area	Plicae circulares (folds) × villi (finger-like projections) × microvilli (brush border) = ~250 m^2. Form follows function: huge surface to absorb every nutrient before food exits.
Pancreatic juice: enzymes + buffer	Trypsinogen, chymotrypsinogen, carboxypeptidase, pancreatic amylase, lipase + bicarbonate to neutralize gastric acid.
Bile: where made, where stored, what it does	Made in liver. Stored in gallbladder. Released into duodenum (CCK trigger). Emulsifies fat into micelles → lipase can act on it. Without bile, fats are not absorbed.
Fat-soluble vitamins (and why they fail in malabsorption)	A, D, E, K. They need bile + intact villi to be absorbed. Lose the villi (Juan) and you lose all four. Vitamin D loss → calcium malabsorption → osteopenia/osteoporosis.
Glucose absorption mechanism	Apical: SGLT1 (Na^+/glucose secondary active transport). Basolateral: GLUT2 (facilitated diffusion into blood). Na^+/K^+ ATPase keeps the gradient.

> **Juan (celiac).** Anti-gluten autoimmune attack flattens villi. Lose surface area → malabsorb everything, especially fat-soluble vitamins. Long term: D ↓ → Ca^{2+} ↓ → bone disease. *One villus → one bone.*

Unit 2 · Metabolism & Nutrition (Week 2)

Three stages of aerobic catabolism (and where each happens)	**Glycolysis:** cytoplasm. 1 glucose → 2 pyruvate, net 2 ATP, 2 NADH. **Citric acid cycle:** mitochondrial matrix. 2 ATP, 6 NADH, 2 FADH&sub2; per glucose. **ETC + chemiosmosis:** inner mitochondrial membrane. ~32 ATP. Total ~36 ATP per glucose.
What chemiosmosis means	ETC pumps H^+ from matrix into intermembrane space → gradient. ATP synthase lets H^+ back into matrix → spins → phosphorylates ADP → ATP. Same logic in mitochondria as in chloroplasts.
Insulin: 4 things it does	Glucose into cells (GLUT4), glycogen synthesis, fat storage, protein synthesis. *Anabolic hormone of the fed state.*

Glucagon: 4 things it does	Glycogenolysis, gluconeogenesis, lipolysis, ketogenesis. *Catabolic hormone of the fasted state.*
Gluconeogenesis substrates	Glycerol (from fat), lactate (Cori cycle), gluconeogenic amino acids (mostly alanine). *Liver makes glucose from non-carb carbon when glycogen runs out (~24 h fast).*
In starvation, what fuel does the brain switch to?	Initially: glucose from gluconeogenesis. After several days: ketone bodies (from liver fatty-acid oxidation). Spares muscle protein.
Macronutrients and their kcal/g	Carbs: 4 kcal/g. Protein: 4 kcal/g. Fat: 9 kcal/g. Alcohol: 7 kcal/g.
Essential nutrients (cannot be synthesized)	9 essential amino acids, 2 essential fatty acids (linoleic, linolenic), all vitamins, all minerals.

> **Crystal & the Brown Bag patients.** Anabolic vs. catabolic switching is one mechanism (insulin/glucagon ratio). Same building blocks become muscle, glycogen, or fat depending on context. Hyperglycemia = the cell selecting the wrong fuel because the hormone signal is broken.

Unit 3 · Urinary System (Week 3)

Five regions of the nephron	Glomerulus (filter) → PCT (bulk reabsorption) → loop of Henle (countercurrent, medullary gradient) → DCT (fine-tuning, aldosterone-sensitive) → collecting duct (ADH, principal/intercalated cells).
Three layers of the glomerular filter (in order)	Fenestrated endothelium → basal lamina (negative charge, blocks albumin) → podocytes (filtration slits). Damage any layer = proteinuria.
GFR equation in plain language	GFR = ($P_{\text{glomerular capillary}}$) - ($P_{\text{Bowman's space}} + \pi_{\text{glomerular capillary}}$). Pressure pushing out vs. pressures pushing back. Normal ~125 mL/min.
Where does most reabsorption happen?	PCT. ~65% of water + Na^+ reabsorbed there, obligatory, isosmotic. Glucose, amino acids, bicarbonate fully reabsorbed if normal.
Loop of Henle: what for?	Sets up the medullary osmotic gradient ($Na^+/K^+/2Cl^-$ transporter in thick ascending limb). The gradient is what lets the collecting duct concentrate urine when ADH is on.
ADH: action, receptor, target	V2 receptor on principal cells of collecting duct → inserts aquaporin-2 → water reabsorbed down the medullary gradient → concentrated urine, dilute plasma.
Aldosterone: action, target	Mineralocorticoid receptor on principal cells of collecting duct → upregulates ENaC, Na^+/K^+ ATPase, ROMK → reabsorbs Na^+, secretes K^+ and H^+. *Aldosterone exchanges sodium IN for potassium and hydrogen OUT.*
RAAS cascade in 5 steps	Low BP → juxtaglomerular cells release renin → renin cleaves angiotensinogen to angiotensin I → ACE (lung) converts to angiotensin II → AT-II constricts efferent arteriole + triggers aldosterone + thirst.
Why ACE inhibitors lower GFR	No AT-II → efferent arteriole dilates → pressure in glomerular capillary drops → GFR drops. (Protective in diabetic nephropathy because it reduces glomerular pressure.)
Renal clearance: why creatinine is used as a GFR proxy	Filtered freely, not reabsorbed, only mildly secreted (~10% overestimate). So clearance of creatinine ~ GFR.

> **Santiago (HUS).** Toxin from E. coli O157:H7 destroys endothelium of glomerular capillaries. Filter fails → proteinuria, hematuria, acute

> kidney injury. The kidney cannot excrete H^+ or generate new HCO_3^- → uremic acidosis (the U in MUDPILES).

Unit 4 · Fluid, Electrolyte & Acid-Base Balance (Week 4)

Body fluid compartments by volume	Total body water ~42 L (~60% body weight). ICF = 2/3 (~28 L). ECF = 1/3 (~14 L), of which plasma is ~3 L and interstitial ~11 L.
Tonicity: what it controls	Whether a cell shrinks (hypertonic), swells (hypotonic), or stays the same (isotonic). Driven by impermeable solutes (mostly Na^+ for ECF).
Starling forces (and Reyna's edema)	Net filtration = (Pc - Pi) - σ(πc - πi). Reyna loses albumin in urine → πc falls → fluid leaks into interstitium → leg edema.
ADH trigger: primary vs. secondary	Primary = high plasma osmolality, sensed by hypothalamic osmoreceptors (very sensitive, threshold ~280). Secondary = severe volume loss sensed by baroreceptors (only kicks in at >10% volume drop).
Aldosterone trigger: primary vs. secondary	Primary = angiotensin II (downstream of RAAS, low perfusion). Secondary = high plasma K^+ directly on the zona glomerulosa.
ANP: trigger and effect	Trigger: atrial stretch (volume overload). Effect: opposite of RAAS — dumps Na^+ and water, dilates afferent arteriole, suppresses renin/aldosterone.
IV fluid choice: NS, D5W, LR	**NS (0.9% NaCl):** ECF volume expansion (resuscitation). **D5W:** free water replacement (hypernatremia, dehydration). **LR:** balanced resuscitation, has K^+ (avoid in hyperkalemia/CKD).
Henderson-Hasselbalch in plain words	pH depends on the *ratio* of HCO_3^- to PCO_2, not their absolute values. Normal ratio = 20:1. Lungs control the denominator (PCO_2); kidneys control the numerator (HCO_3^-).
Four primary acid-base disturbances	Respiratory acidosis (↑PCO_2), respiratory alkalosis (↓PCO_2), metabolic acidosis (↓HCO_3^-), metabolic alkalosis (↑HCO_3^-).
30-second ABG algorithm	(1) Is pH acidic or alkalotic? (2) Does PCO_2 go with pH (respiratory) or against it (metabolic primary)? (3) Does HCO_3^- go with pH (metabolic) or against it (respiratory primary)? (4) Compensation? Lungs partial/fast, kidneys complete/slow.
Anion gap: formula and meaning	AG = Na^+ - (Cl^- + HCO_3^-). Normal 8-12. High AG (HAGMA) = an organic acid is being added (DKA, lactate). Normal AG (NAGMA) = bicarbonate is being lost (diarrhea, RTA).
MUDPILES (HAGMA causes)	**M**ethanol, **U**remia, **D**KA, **P**ropylene glycol, **I**NH/**I**ron, **L**actic acidosis, **E**thylene glycol, **S**alicylates.
Compensation timescales	Buffer (instant, exhaustible). Lungs (minutes, partial). Kidneys (hours-to-days, complete). Acute respiratory disturbances are NOT yet renally compensated; chronic ones are.

> **Reyna (diabetic nephropathy).** Slow glomerular damage from chronic hyperglycemia. Edema (low oncotic from albuminuria) + hypertension (RAAS over-active because kidney perceives low perfusion). First-line: ACE inhibitor — protective even when GFR is normal.

> **Oscar (primary aldosteronism / Conn's).** Adrenal autonomously over-secretes aldosterone → *one hormone explains four labs:* high BP (Na^+/water retention), low K^+ (kaliuresis), high pH (H^+ secretion drives metabolic alkalosis), suppressed renin (negative feedback from high BP). Treat: spironolactone (K^+-sparing) or surgery.

Integrative Themes (these come up everywhere)

> **Three sensors.** *Osmoreceptors* (osmolality, hypothalamus, drive ADH/thirst). *Baroreceptors* (volume/pressure, carotid + aortic + JG cells, drive sympathetic + RAAS). *Chemoreceptors* ($pH/CO_2/O_2$, peripheral + central, drive ventilation).

> **Three timescales.** Buffer (instant, milliseconds). Lung (minutes, fast, partial). Kidney (hours-to-days, slow, complete). Same hierarchy applies to fluid balance: sympathetic (seconds) → ADH/RAAS (minutes-to-hours) → aldosterone-driven structural change (days).

> **Three knobs of fluid & Na^+ balance.** ADH = water IN. Aldosterone = Na^+ IN, K^+/H^+ OUT. ANP = both OUT. Predict any fluid disorder by asking which knob is stuck and which way.

> **Form follows function.** Villi/microvilli for surface area. Three filter layers in the glomerulus for size + charge selectivity. Counter-current loop architecture for the medullary gradient. The shape of the structure tells you what it does.

What to Re-do This Weekend (in this order)

1. **Each patient case from a blank sheet of paper, from memory.** Juan, Crystal, the Brown Bag patients, Santiago, Reyna, Oscar. For each: signs → cellular mechanism → treatment. If you can rebuild the case, you have it.

2. **The 30-second ABG algorithm on three made-up examples.** Pick any $pH/PCO_2/HCO_3^-$ trio in your head, classify it, predict compensation.

3. **Draw the nephron and the three filter layers.** Annotate where each diuretic class acts.

4. **Write the four-cell ABG matrix from memory and put one example in each cell.**

5. **Sketch the bicarbonate equation** with the lungs sitting on the CO_2 side and the kidneys sitting on the HCO_3^- side. Predict what happens to pH if either side moves.

If you can do all five of these cold by Tuesday night, you are ready for Wednesday. See you Monday for the practice. — Giorgio